NYSC REGISTRATION DATASET

SQL Practice Exercises — Questions, Queries & Findings

MySQL Workbench walkthrough + presentation-ready answers

Dataset: 10,000 registrants • 11 columns

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Executive Summary

PROJECT AT A GLANCE

REGISTRANTS	AVERAGE AGE	INSTITUTIONS	COURSES
10,000	**24.96**	**10**	**15**
Records analyzed	Age range: 20–30	Distinct institutions	Distinct courses

- The dataset contains 10,000 registrants and has no missing values in the 11 supplied columns.
- The most common course is Biology (711 registrants).
- Cross River is the leading institution state by registrant count (1,049).
- SQL techniques used include filtering, sorting, aggregation, subqueries, and window functions.

Project Overview & Objectives

INTRODUCTION

PROJECT OVERVIEW

This project uses SQL to explore NYSC registration records and summarize registrants by age, origin state, preferred service state, institution, course of study, and graduating honours.

OBJECTIVES

- Retrieve selected registrant details and filter records using conditions.
- Identify distributions and counts by state, institution, course, and honours.
- Use subqueries to answer comparative and ranking questions.
- Summarize actionable findings for NYSC planning and reporting.

PROJECT GOAL

Turn raw registration records into clear, reproducible insights using MySQL queries.

Dataset Profile & Workflow

DATA UNDERSTANDING

ROWS	COLUMNS	AGE RANGE
10,000 records	11 fields	20–30 years

MISSING VALUES	INSTITUTION COUNT	COURSE COUNT
0 across supplied columns	10	15

WORKFLOW

- Import CSV
- Check columns
- Run SQL queries
- Review result grids
- Visualize findings

Key fields: Fullname, Age, Origin State, Preferred State, Institution, Institution State, Course Study, Graduating Honors, and contact fields.

DATA NOTE

The supplied dataset does not include a Gender column; gender-based age comparisons require verified gender data.

SECTION 1 — Basic SELECT & Filtering

Question and answer — use the query in MySQL Workbench

1. Retrieve all columns from the table.

```
SELECT *
FROM `nysc_registration_data`;
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
1	Norma Fisher	26	normafisher6891@gmail.com	Kano	Anambra	University of Calabar	Cross River	Economics	876-475-9382x421	Third Class
2	Danielle Browning	24	daniellebrowning7809@gmail.com	Sokoto	Edo	University of Ibadan	Oyo	Political Science	001-924-115-7815x659	Second Class Upper
3	Christine Tran	24	christinetran2290@gmail.com	Rivers	Bauchi	Obafemi Awolowo University	Osun	Biology	840.801.6097x535	Pass
4	Amy Stark	29	amystark2408@gmail.com	Borno	Rivers	Ahmadu Bello University	Kaduna	Biology	328.711.5871	Second Class Lower
5	Eddie Martinez	27	eddiemartinez9172@gmail.com	Rivers	Sokoto	University of Port Harcourt	Rivers	Medicine	839-894-7196x59342	Pass
6	Cody Salazar	30	codysalazar3351@gmail.com	Cross River	Osun	Bayero University	Kano	Chemistry	+1-471-122-0186x84833	Pass
7	Leslie Herring	24	leslieherring1021@gmail.com	Cross River	Lagos	Ahmadu Bello University	Kaduna	Mathematics	001-477-515-9179	Third Class
8	Sarah Sanchez	30	sarahsanchez19@gmail.com	Bauchi	Osun	University of Benin	Edo	Business Administration	001-413-525-6012x3098	Second Class Lower
9	Angela Martinez	21	angelamartinez3131@gmail.com	Edo	Delta	University of Ibadan	Oyo	Physics	991.615.1090	Second Class Upper
10	Wesley Robbins	28	wesleyrobbins7340@gmail.com	Kaduna	Kaduna	University of Benin	Edo	Biology	300.869.1413x1456	Pass
11	Jonathan Williams	27	jonathanwilliams1787@gmail.com	Borno	Cross River	Obafemi Awolowo University	Osun	Mathematics	634-579-2302	First Class
12	Gloria King	28	gloriaking5452@gmail.com	Cross River	Enugu	Abubakar Tafawa Balewa U...	Bauchi	Political Science	+1-720-769-8456x428	Pass
13	Judith Miller	24	judithmiller7291@gmail.com	Kaduna	Bauchi	University of Port Harcourt	Rivers	Medicine	001-084-237-5945x9924	Pass
14	Jason Curtis	23	jasoncurtis4757@gmail.com	Oyo	Enugu	University of Nigeria	Enugu	Computer Science	9352337696	Pass
15	Shane Bradley	30	shanebradley4261@gmail.com	Osun	Kaduna	Ahmadu Bello University	Kaduna	Agriculture	001-027-142-7878x90075	Second Class Upper

Answer / interpretation: Returns all columns and all 10,000 records.

SECTION 1 — Basic SELECT & Filtering

Question and answer — use the query in MySQL Workbench

2. Show only the Fullname, Age, and Gmail of all registrants.

```
SELECT `Fullname`, `Age`, `Gmail`
FROM `nysc_registration_data`;
```

Fullname	Age	Gmail
Norma Fisher	26	normafisher6891@gmail.com
Danielle Browning	24	daniellebrowning7809@gmail.com
Christine Tran	24	christinetran2290@gmail.com
Amy Stark	29	amystark2408@gmail.com
Eddie Martinez	27	eddiemartinez9172@gmail.com
Cody Salazar	30	codysalazar3351@gmail.com
Leslie Herring	24	leslieherring1021@gmail.com
Sarah Sanchez	30	sarahsanchez19@gmail.com
Angela Martinez	21	angelamartinez3131@gmail.com
Wesley Robbins	28	wesleyrobbins7340@gmail.com
Jonathan Williams	27	jonathanwilliams1787@gmail.com
Gloria King	28	gloriaking5452@gmail.com
Judith Miller	24	judithmiller7291@gmail.com
Jason Curtis	23	jasoncurtis4757@gmail.com
Shane Bradley	30	shanebradley4261@gmail.com
John Lamb	22	johnlamb634@gmail.com
Lee Gonzalez	24	leegonzalez8550@gmail.com

Answer / interpretation: Returns the three requested columns for all registrants.

SECTION 1 — Basic SELECT & Filtering

Question and answer — use the query in MySQL Workbench

3. List all registrants whose age is greater than 25.

```
SELECT *
FROM `nysc_registration_data`
WHERE `Age` > 25
LIMIT 15;
```

Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
Norma Fisher	26	normafisher6891@gmail.com	Kano	Anambra	University of Calabar	Cross River	Economics	876-475-9382x421	Third Class
Amy Stark	29	amystark2408@gmail.com	Borno	Rivers	Ahmadu Bello University	Kaduna	Biology	328.711.5871	Second Class Lower
Eddie Martinez	27	eddiemartinez9172@gmail.com	Rivers	Sokoto	University of Port Harcourt	Rivers	Medicine	839-894-7196x59342	Pass
Cody Salazar	30	codysalazar3351@gmail.com	Cross River	Osun	Bayero University	Kano	Chemistry	+1-471-122-0186x84833	Pass
Sarah Sanchez	30	sarahsanchez19@gmail.com	Bauchi	Osun	University of Benin	Edo	Business Administration	001-413-525-6012x3098	Second Class Lower
Wesley Robbins	28	wesleyrobbins7340@gmail.com	Kaduna	Kaduna	University of Benin	Edo	Biology	300.869.1413x1456	Pass
Jonathan Williams	27	jonathanwilliams1787@gmail.com	Borno	Cross River	Obafemi Awolowo University	Osun	Mathematics	634-579-2302	First Class
Gloria King	28	gloriaking5452@gmail.com	Cross River	Enugu	Abubakar Tafawa Balewa University	Bauchi	Political Science	+1-720-769-8456x428	Pass
Shane Bradley	30	shanebradley4261@gmail.com	Osun	Kaduna	Ahmadu Bello University	Kaduna	Agriculture	001-027-142-7878x90075	Second Class Upper
Christopher York	29	christopheryork4510@gmail.com	Plateau	Osun	University of Benin	Edo	Electrical Engineering	0348559097	Second Class Lower
Stephanie Bowman	29	stephaniebowman1890@gmail.com	Osun	Edo	University of Benin	Edo	Chemistry	(823)694-0224x555	Second Class Upper
Amy Smith	30	amysmith3089@gmail.com	Bauchi	Edo	Ahmadu Bello University	Kaduna	Pharmacy	(884)779-3615x3492	First Class
Yvette Reid	29	yvettereid1663@gmail.com	Anambra	Kaduna	University of Ibadan	Oyo	Electrical Engineering	893.055.5082	Second Class Lower
Christopher Hern...	29	christopherhernandez1654@gm...	Zamfara	Enugu	Obafemi Awolowo University	Osun	Medicine	+1-758-688-0918x9163	Third Class
Steven Butler	29	stevenbutler2776@gmail.com	Enugu	Kano	University of Nigeria	Enugu	Chemistry	699.300.2489x4517	Second Class Upper

Answer / interpretation: Returns records for registrants aged 26–30.

SECTION 1 — Basic SELECT & Filtering

Question and answer — use the query in MySQL Workbench

4. Display all records where the origin state is 'Lagos'.

```
SELECT *
FROM `nysc_registration_data`
WHERE `Origin State` = 'Lagos';
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
31	Megan Hernandez	30	meganhernandez2865@gmail.com	Lagos	Osun	University of Port Harcourt	Rivers	Biology	(609)767-0172x009	Pass
45	Gerald Moreno	30	geraldmoreno1333@gmail.com	Lagos	Anambra	Bayero University	Kano	Physics	216.448.2983	First Class
97	Jacob Thomas	30	jacobthomas2019@gmail.com	Lagos	Borno	University of Benin	Edo	Mathematics	(780)342-9546x048	First Class
127	Diana Roberson	27	dianaroberson6406@gmail.com	Lagos	Ekiti	Ahmadu Bello University	Kaduna	Agriculture	668.539.4102	First Class
128	Kevin Washington	30	kevinwashington6474@gmail.com	Lagos	Sokoto	University of Lagos	Lagos	Electrical Engineering	765.601.6922x01630	Pass
147	Kathy Willis	28	kathywillis1186@gmail.com	Lagos	Plateau	Bayero University	Kano	Mathematics	006.902.8164	Third Class
174	Patricia Marquez	26	patriciamarquez6270@gmail.com	Lagos	Katsina	Obafemi Awolowo University	Osun	Political Science	001-204-978-9637	Pass
180	Jessica Swanson	25	jessicaswanson6169@gmail.com	Lagos	Kano	Bayero University	Kano	Electrical Engineering	(602)714-2839x7529	Second Class Lower
199	Blake Johnson	26	blakejohnson3741@gmail.com	Lagos	Sokoto	University of Calabar	Cross River	Mechanical Engineering	503.946.7841x0156	Second Class Upper
207	Jill Mason	24	jillmason5376@gmail.com	Lagos	Zamfara	Bayero University	Kano	Biology	(812)568-2180	Pass
222	Mark Hart	22	markhart2050@gmail.com	Lagos	Benue	Abubakar Tafawa Balewa U...	Bauchi	Business Administration	001-414-527-5051	First Class
227	Caitlin Benjamin	21	caitlinbenjamin2022@gmail.com	Lagos	Sokoto	University of Lagos	Lagos	Mechanical Engineering	841-061-6772	Third Class
325	Justin Johnson	24	justinjohnson1372@gmail.com	Lagos	Delta	Abubakar Tafawa Balewa U...	Bauchi	Chemistry	(442)197-8593x80378	First Class
379	Rick Campbell	20	rickcampbell2822@gmail.com	Lagos	Bauchi	University of Calabar	Cross River	Mass Communication	6023413097	Second Class Lower
382	Michael Williams	30	michaelwilliams3715@gmail.com	Lagos	Kano	University of Port Harcourt	Rivers	Biology	591.864.3277	Pass

Answer / interpretation: Filters records to registrants whose origin state is Lagos.

SECTION 1 — Basic SELECT & Filtering

Question and answer — use the query in MySQL Workbench

5. Find all registrants who graduated with 'First Class'.

```
SELECT *
FROM `nysc_registration_data`
WHERE `Graduating Honors` = 'First Class';
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
11	Jonathan Williams	27	jonathanwilliams1787@gmail.com	Borno	Cross River	Obafemi Awolowo University	Osun	Mathematics	634-579-2302	First Class
22	Vanessa Cooper	20	vanessacooper9403@gmail.com	Cross River	Bauchi	Ahmadu Bello University	Kaduna	Computer Science	176.045.2296x1113	First Class
23	Amy Smith	30	amysmith3089@gmail.com	Bauchi	Edo	Ahmadu Bello University	Kaduna	Pharmacy	(884)779-3615x3492	First Class
27	Lindsey Moore	25	lindseymoore7145@gmail.com	Oyo	Kano	University of Calabar	Cross River	Economics	(947)118-0132x0407	First Class
35	Steven Henderson	29	stevenhenderson2169@gmail.com	Borno	Zamfara	University of Port Harcourt	Rivers	Chemistry	161-162-0766	First Class
39	Emily Sanchez	22	emilysanchez7314@gmail.com	Ekiti	Osun	University of Calabar	Cross River	Education	001-245-602-9061x12648	First Class
45	Gerald Moreno	30	geraldmoreno1333@gmail.com	Lagos	Anambra	Bayero University	Kano	Physics	216.448.2983	First Class
51	Bradley Lopez	22	bradleylopez4447@gmail.com	Benue	Benue	University of Benin	Edo	Mathematics	2394918456	First Class
53	Dylan Smith	24	dylansmith5914@gmail.com	Zamfara	Cross River	University of Nigeria	Enugu	Mass Communication	639-683-7359	First Class
55	Nicholas Turner	21	nicholasturner4959@gmail.com	Zamfara	Benue	Obafemi Awolowo University	Osun	Pharmacy	647.530.4810x77706	First Class
58	Lisa Mccormick	25	lisamccormick2551@gmail.com	Oyo	Edo	University of Port Harcourt	Rivers	Physics	001-478-978-4430x191	First Class
59	Jason Robinson	21	jasonrobinson1388@gmail.com	Enugu	Delta	University of Lagos	Lagos	Pharmacy	001-088-601-6419x5384	First Class
62	Olivia Gomez	29	oliviagomez2730@gmail.com	Katsina	Enugu	University of Benin	Edo	Electrical Engineering	001-271-908-9577	First Class
64	Rachel Cruz	24	rachelcruz3396@gmail.com	Kano	Enugu	Abubakar Tafawa Balewa U...	Bauchi	Mechanical Enginee...	3053573452	First Class
70	Johnathan Long	26	johnathanlong9552@gmail.com	Borno	Osun	Ahmadu Bello University	Kaduna	Physics	848-708-9840x7616	First Class

Answer / interpretation: Returns all records with First Class honors.

SECTION 1 — Basic SELECT & Filtering

Question and answer — use the query in MySQL Workbench

6. Retrieve the names of registrants whose preferred state is 'Abuja' and origin state is 'Kano'.

```
SELECT `Fullname`
FROM `nysc_registration_data`
WHERE `Preferred State` = 'Abuja'
  AND `Origin State` = 'Kano';
```

Fullname	Preferred State	Origin State
Lauren Wood	Abuja	Kano
Zachary Smith	Abuja	Kano
Courtney Smith	Abuja	Kano
Connie Higgins	Abuja	Kano
Jennifer Bell	Abuja	Kano
Nathan Williams	Abuja	Kano
Edward Cunningham	Abuja	Kano
Kevin Robinson	Abuja	Kano
Christian Harvey	Abuja	Kano
Justin Sosa	Abuja	Kano
Frederick Bailey	Abuja	Kano
Kimberly Williams	Abuja	Kano
Lauren Wells	Abuja	Kano
Sharon Moore	Abuja	Kano
John Watson	Abuja	Kano
David Rodriguez	Abuja	Kano

Answer / interpretation: Returns names satisfying both conditions.

SECTION 1 — Basic SELECT & Filtering

Question and answer — use the query in MySQL Workbench

7. Show all registrants whose email address ends with '@gmail.com'.

```
SELECT *
FROM `nysc_registration_data`
WHERE `Gmail` LIKE '%@gmail.com';
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
1	Norma Fisher	26	normafisher6891@gmail.com	Kano	Anambra	University of Calabar	Cross River	Economics	876-475-9382x421	Third Class
2	Danielle Browning	24	daniellebrowning7809@gmail.com	Sokoto	Edo	University of Ibadan	Oyo	Political Science	001-924-115-7815x659	Second Class Uppe
3	Christine Tran	24	christinetran2290@gmail.com	Rivers	Bauchi	Obafemi Awolowo University	Osun	Biology	840.801.6097x535	Pass
4	Amy Stark	29	amystark2408@gmail.com	Borno	Rivers	Ahmadu Bello University	Kaduna	Biology	328.711.5871	Second Class Lowe
5	Eddie Martinez	27	eddiemartinez9172@gmail.com	Rivers	Sokoto	University of Port Harcourt	Rivers	Medicine	839-894-7196x59342	Pass
6	Cody Salazar	30	codysalazar3351@gmail.com	Cross River	Osun	Bayero University	Kano	Chemistry	+1-471-122-0186x84833	Pass
7	Leslie Herring	24	leslieherring1021@gmail.com	Cross River	Lagos	Ahmadu Bello University	Kaduna	Mathematics	001-477-515-9179	Third Class
8	Sarah Sanchez	30	sarahsanchez19@gmail.com	Bauchi	Osun	University of Benin	Edo	Business Administration	001-413-525-6012x3098	Second Class Lowe
9	Angela Martinez	21	angelamartinez3131@gmail.com	Edo	Delta	University of Ibadan	Oyo	Physics	991.615.1090	Second Class Uppe
10	Wesley Robbins	28	wesleyrobbins7340@gmail.com	Kaduna	Kaduna	University of Benin	Edo	Biology	300.869.1413x1456	Pass
11	Jonathan Williams	27	jonathanwilliams1787@gmail.com	Borno	Cross River	Obafemi Awolowo University	Osun	Mathematics	634-579-2302	First Class
12	Gloria King	28	gloriaking5452@gmail.com	Cross River	Enugu	Abubakar Tafawa Balewa U...	Bauchi	Political Science	+1-720-769-8456x428	Pass
13	Judith Miller	24	judithmiller7291@gmail.com	Kaduna	Bauchi	University of Port Harcourt	Rivers	Medicine	001-084-237-5945x9924	Pass
14	Jason Curtis	23	jasoncurtis4757@gmail.com	Oyo	Enugu	University of Nigeria	Enugu	Computer Science	9352337696	Pass
15	Shane Bradley	30	shanebradley4261@gmail.com	Osun	Kaduna	Ahmadu Bello University	Kaduna	Agriculture	001-027-142-7878x90075	Second Class Uppe

Answer / interpretation: Returns rows whose Gmail field ends with @gmail.com.

SECTION 2 — Sorting & Distinct

Question and answer — use the query in MySQL Workbench

1. List all unique origin states in the dataset.

```
SELECT DISTINCT `Origin State`
FROM `nysc_registration_data`
ORDER BY `Origin State`;
```

Origin State
Abuja
Anambra
Bauchi
Benue
Borno
Cross River
Delta
Edo
Ekiti
Enugu
Kaduna
Kano
Katsina
Kano
Katsina
Lagos
Osun
Oyo
Plateau
Rivers
Sokoto
Zamfara

Answer / interpretation: The dataset contains 20 distinct origin states.

SECTION 2 — Sorting & Distinct

Question and answer — use the query in MySQL Workbench

2. Display all registrants ordered by Age in descending order.

```
SELECT *
FROM `nysc_registration_data`
ORDER BY `Age` DESC;
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
9438	Elizabeth Anderson	30	elizabethanderson3621@gmail.com	Zamfara	Kano	Obafemi Awolowo University	Osun	Pharmacy	3696111041	Pass
9461	Tyler Frazier	30	tylerfrazier5769@gmail.com	Anambra	Delta	Ahmadu Bello University	Kaduna	Chemistry	721-602-9640x8342	Second Class Upper
9474	Robert Joseph	30	robertjoseph1798@gmail.com	Osun	Enugu	University of Calabar	Cross River	Computer Science	1037885004	Pass
9503	Joy Welch	30	joywelch9778@gmail.com	Plateau	Kano	University of Ibadan	Oyo	Business Administration	680-420-2116	Third Class
6294	James Burke	29	jamesburke9385@gmail.com	Benue	Ekiti	University of Benin	Edo	Physics	+1-134-006-2007x73975	Second Class Lower
6295	Jennifer Rodriguez	29	jenniferrodriguez7238@gmail.com	Katsina	Lagos	Bayero University	Kano	Electrical Engineering	270.759.6313x25193	Second Class Lower
6329	Warren Jones	29	warrenjones2719@gmail.com	Osun	Katsina	University of Benin	Edo	Chemistry	907-778-8328x6624	Pass
6330	Justin Barrett	29	justinbarrett5236@gmail.com	Zamfara	Sokoto	University of Calabar	Cross River	Business Administration	(951)166-2682x0516	Pass
6335	Kelsey Roberts	29	kelseyroberts9570@gmail.com	Rivers	Cross River	Ahmadu Bello University	Kaduna	Business Administration	418-590-8105	Second Class Lower
7921	Sheila Singleton	28	sheilasingleton704@gmail.com	Enugu	Anambra	University of Lagos	Lagos	Electrical Engineering	280.678.4372x376	Second Class Upper
7922	Joshua Rogers	28	joshuarogers3946@gmail.com	Osun	Katsina	University of Port Harcourt	Rivers	Pharmacy	(750)888-5883	Second Class Lower
7950	Beth Brown	28	bethbrown1261@gmail.com	Abuja	Kaduna	Abubakar Tafawa Balewa U...	Bauchi	Mechanical Engineering	340-794-2104x164	First Class
7956	Sean Williams	28	seanwilliams1765@gmail.com	Lagos	Plateau	Bayero University	Kano	Agriculture	001-897-077-0088x995	First Class
7958	Tyler Mccoy	28	tylermccoy4060@gmail.com	Abuja	Osun	University of Port Harcourt	Rivers	Mathematics	001-509-001-0051	Pass
7976	Timothy Smith	28	timothysmith7070@gmail.com	Osun	Sokoto	Bayero University	Kano	Electrical Engineering	253-964-9891x4600	Third Class
5278	Timothy Bradford	27	timothybradford3613@gmail.com	Cross River	Rivers	Obafemi Awolowo University	Osun	Electrical Engineering	599.471.3248x998	Pass
5280	Tracy Lopez	27	tracylopez7309@gmail.com	Sokoto	Abuja	University of Nigeria	Enugu	Pharmacy	081.866.1150x277	Second Class Upper
5292	David Smith	27	davidsmith5207@gmail.com	Lagos	Delta	Abubakar Tafawa Balewa U...	Bauchi	Mass Communication	740-727-1402	Second Class Upper
7848	Zachary Ellis	26	zacharyellis3221@gmail.com	Plateau	Sokoto	Abubakar Tafawa Balewa U...	Bauchi	Political Science	986.089.3476x51858	Third Class
7854	Phillip Maldonado	26	phillipmaldonado2948@gmail.com	Katsina	Kano	University of Ibadan	Oyo	Physics	317-211-7275x9553	Pass
1	Norma Fisher	26	normafisher6891@gmail.com	Kano	Anambra	University of Calabar	Cross River	Economics	876-475-9382x421	Third Class
7576	Amber Rosales	20	amberrosales9349@gmail.com	Oyo	Osun	University of Nigeria	Enugu	Biology	001-989-194-9488x0557	Second Class Lower
7601	Susan Gay	20	susangay811@gmail.com	Plateau	Bauchi	Abubakar Tafawa Balewa U...	Bauchi	Biology	8388911108	First Class
7604	Rachel Padilla	20	rachelpadilla8398@gmail.com	Borno	Kaduna	Bayero University	Kano	Economics	+1-271-323-3862x287	Second Class Upper
7610	Jose Joseph	20	josejoseph825@gmail.com	Rivers	Oyo	Obafemi Awolowo University	Osun	Medicine	+1-359-843-7021x601	First Class
7619	Susan Santiago	20	susansantiago4373@gmail.com	Rivers	Plateau	University of Lagos	Lagos	Political Science	(558)270-8079x8913	Second Class Upper

Answer / interpretation: Sorts oldest to youngest; equal ages may appear in any order.

SECTION 2 — Sorting & Distinct

Question and answer — use the query in MySQL Workbench

3. Show the top 5 oldest registrants.

```
SELECT *
FROM `nysc_registration_data`
ORDER BY `Age` DESC, `Fullname` ASC
LIMIT 5;
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
2055	Adriana Ramirez	30	adrianaramirez2273@gmail.com	Anambra	Kaduna	University of Nigeria	Enugu	Chemistry	+1-204-707-4547	Second Class Lower
6702	Alan Hunter	30	alanhunter450@gmail.com	Anambra	Kaduna	Abubakar Tafawa Balewa University	Bauchi	Education	7390263751	Second Class Upper
471	Albert Martin	30	albertmartin9522@gmail.com	Ekiti	Oyo	Bayero University	Kano	Agriculture	001-839-453-6586x556	Second Class Upper
4788	Alejandro Hill	30	alejandrohill4444@gmail.com	Zamfara	Kaduna	University of Ibadan	Oyo	Education	001-706-424-2930x31099	Pass
4702	Alejandro Jackson	30	alejandrojackson2051@gmail.com	Plateau	Edo	Ahmadu Bello University	Kaduna	Physics	001-027-684-0433x7052	Third Class

Answer / interpretation: Returns five rows at the maximum age, 30, with a stable tie-breaker.

SECTION 2 — Sorting & Distinct

Question and answer — use the query in MySQL Workbench

4. Find all unique institutions in the dataset.

```
SELECT DISTINCT `Institution`
FROM `nysc_registration_data`
ORDER BY `Institution`;
```

Institution
Abubakar Tafawa Balewa University
Ahmadu Bello University
Bayero University
Obafemi Awolowo University
University of Benin
University of Calabar
University of Ibadan
University of Lagos
University of Nigeria
University of Port Harcourt

Answer / interpretation: The dataset contains 10 distinct institutions.

SECTION 3 — Aggregation & Grouping

Question and answer — use the query in MySQL Workbench

1. Count the total number of registrants.

```sql
SELECT FORMAT(COUNT(Id), 'N2') AS `Total Registrants`
FROM `nysc_registration_data`;
```

Total Registrants
10,000

Answer / interpretation: Result: 10,000 registrants.

SECTION 3 — Aggregation & Grouping

Question and answer — use the query in MySQL Workbench

2. Find the average age of all registrants.

```
SELECT ROUND(AVG(`Age`), 0) AS
        `Average Age`
FROM `nysc_registration_data`;
```

Average Age
25

Answer / interpretation: Result: 25 years (rounded to nearest whole number).

SECTION 3 — Aggregation & Grouping

Question and answer — use the query in MySQL Workbench

3. Count how many registrants come from each origin state.

```
SELECT `Origin State`, COUNT(*) AS `Total Registrants`
FROM `nysc_registration_data`
GROUP BY `Origin State`
ORDER BY `Total Registrants` DESC;
```

Origin State	Total Registrants
Borno	541
Benue	532
Katsina	525
Bauchi	517
Sokoto	516
Abuja	514
Enugu	511
Edo	507
Ekiti	507
Zamfara	499
Lagos	495
Kano	493
Osun	493
Rivers	491
Anambra	488
Oyo	486
Kaduna	484
Plateau	483
Cross River	463
Delta	455

Answer / interpretation: One row per origin state, ranked from most to fewest registrants.

SECTION 3 — Aggregation & Grouping

Question and answer — use the query in MySQL Workbench

4. Show the number of registrants from each preferred state, ordered from most to least preferred.

```
SELECT `Preferred State`, COUNT(*) AS
`Registrant Count`
FROM `nysc_registration_data`
GROUP BY `Preferred State`
ORDER BY `Registrant Count` DESC, `Preferred
State` ASC;
```

Preferred State	Registrant Count
Osun	541
Cross River	533
Sokoto	531
Enugu	517
Kano	517
Katsina	514
Rivers	513
Anambra	512
Delta	511
Edo	510
Kaduna	502
Benue	501
Abuja	496
Borno	491
Ekiti	490
Lagos	481
Oyo	481
Bauchi	480
Zamfara	459
Plateau	420

Answer / interpretation: One row per preferred state, ordered by count descending.

SECTION 3 — Aggregation & Grouping

Question and answer — use the query in MySQL Workbench

5. Display the number of graduates by each graduating honors category.

```
SELECT `Graduating Honors`,
FORMAT(COUNT(*), 'N2') AS `Graduate Count`
FROM `nysc_registration_data`
GROUP BY `Graduating Honors`
ORDER BY `Graduate Count` DESC;
```

Graduating Honors	Graduate Count
Second Class Upper	2,080
Third Class	2,021
First Class	2,020
Second Class Lower	1,973
Pass	1,906

Answer / interpretation: Returns counts for First Class, Second Class Upper, Second Class Lower, Third Class, and Pass.

SECTION 3 — Aggregation & Grouping

Question and answer — use the query in MySQL Workbench

6. Find the average age of registrants for each institution.

```
SELECT `Institution`, ROUND(AVG(`Age`), 0)
AS `Average Age`
FROM `nysc_registration_data`
GROUP BY `Institution`
ORDER BY `Average Age` DESC;
```

Institution	Average Age
University of Calabar	25
University of Ibadan	25
Obafemi Awolowo University	25
Ahmadu Bello University	25
University of Port Harcourt	25
Bayero University	25
University of Benin	25
Abubakar Tafawa Balewa University	25
University of Nigeria	25
University of Lagos	25

Answer / interpretation: Returns one average age per institution.

SECTION 4 — Conditional & Filtering Aggregates

Question and answer — use the query in MySQL Workbench

1. Find all institutions that have more than 100 registrants.

```sql
SELECT `Institution`, COUNT(*) AS `Registrant Count`
FROM `nysc_registration_data`
GROUP BY `Institution`
HAVING COUNT(*) > 100
ORDER BY `Registrant Count` DESC;
```

Institution	Registrant Count
University of Calabar	1049
Abubakar Tafawa Balewa University	1038
University of Benin	1029
University of Nigeria	1026
University of Ibadan	1018
University of Lagos	1005
Bayero University	1004
Obafemi Awolowo University	949
Ahmadu Bello University	946
University of Port Harcourt	936

Answer / interpretation: All 10 institutions exceed 100 registrants in this dataset.

SECTION 4 — Conditional & Filtering Aggregates

Question and answer — use the query in MySQL Workbench

2. Count how many registrants from Rivers State prefer to serve in Lagos.

```sql
SELECT COUNT(*) AS `Rivers-to-Lagos Registrants`
FROM `nysc_registration_data`
WHERE `Origin State` = 'Rivers'
  AND `Preferred State` = 'Lagos';
```

Rivers-to-Lagos Registrants
24

Answer / interpretation: Result: 24 registrants.

SECTION 4 — Conditional & Filtering Aggregates

Question and answer — use the query in MySQL Workbench

3. Calculate the average age of registrants who graduated with Second Class Upper.

```sql
SELECT ROUND(AVG(`Age`), 2) AS `Average Age`
FROM `nysc_registration_data`
WHERE `Graduating Honors` = 'Second Class Upper';
```

Average Age
25

Answer / interpretation: Result: 25.01 years.

SECTION 4 — Conditional & Filtering Aggregates

Question and answer — use the query in MySQL Workbench

4. List all courses that have more than 50 graduates.

```sql
SELECT `Course Study`, COUNT(*) AS `Graduate Count`
FROM `nysc_registration_data`
GROUP BY `Course Study`
HAVING COUNT(*) > 50
ORDER BY `Graduate Count` DESC;
```

Course Study	Graduate Count
Biology	711
Medicine	696
Agriculture	679
Mathematics	677
Pharmacy	675
Business Administration	666
Education	665
Political Science	661
Physics	660
Mechanical Engineering	660
Economics	657
Chemistry	656
Computer Science	646
Mass Communication	646
Electrical Engineering	645

Answer / interpretation: Lists courses with more than 50 registrants; all listed course counts are shown by the query.

SECTION 4 — Conditional & Filtering Aggregates

Question and answer — use the query in MySQL Workbench

5. Find the youngest and oldest registrant.

```
SELECT *
FROM `nysc_registration_data`
WHERE `Age` = (SELECT MIN(`Age`)
      FROM `nysc_registration_data`)
   OR `Age` = (SELECT MAX(`Age`)
      FROM `nysc_registration_data`)
ORDER BY `Age`, `Fullname`;
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
1493	Aaron Freeman	20	aaronfreeman3889@gmail.com	Kaduna	Sokoto	University of Nigeria	Enugu	Economics	001-700-621-9871x22597	Pass
7337	Aaron Jimenez	20	aaronjimenez9416@gmail.com	Cross River	Rivers	University of Ibadan	Oyo	Pharmacy	+1-959-317-9258x69916	Third Class
8529	Aaron Long	20	aaronlong2902@gmail.com	Plateau	Benue	University of Benin	Edo	Mathematics	975-153-1267x2446	Second Class Upper
8149	Aaron Thomas	20	aaronthomas290@gmail.com	Enugu	Oyo	Ahmadu Bello University	Kaduna	Biology	001-833-993-5172x47012	Third Class
1088	Abigail Dalton	20	abigaildalton1537@gmail.com	Lagos	Osun	University of Nigeria	Enugu	Business Administration	001-001-972-4236x88144	Second Class Lower
1991	Adam Green	20	adamgreen4505@gmail.com	Ekiti	Rivers	Ahmadu Bello University	Kaduna	Business Administration	001-421-440-8416x401	Third Class
6468	Adam Hill	20	adamhill8807@gmail.com	Abuja	Cross River	University of Lagos	Lagos	Education	175-256-3330x709	Third Class
2811	Adam Huynh	20	adamhuynh540@gmail.com	Zamfara	Oyo	Ahmadu Bello University	Kaduna	Mathematics	(705)741-2951	Third Class
6320	Adrian Davidson	20	adriandavidson5175@gmail.com	Kaduna	Lagos	University of Calabar	Cross River	Agriculture	(306)286-8987	Third Class
2608	Alan Wilson	20	alanwilson1268@gmail.com	Katsina	Oyo	University of Lagos	Lagos	Mathematics	001-527-165-4258	Second Class Lower
4565	Albert Braun	20	albertbraun1056@gmail.com	Zamfara	Oyo	University of Ibadan	Oyo	Biology	(966)552-1003x84250	Pass
2055	Adriana Ramirez	30	adrianaramirez2273@gmail.com	Anambra	Kaduna	University of Nigeria	Enugu	Chemistry	+1-204-707-4547	Second Class Lower
6702	Alan Hunter	30	alanhunter450@gmail.com	Anambra	Kaduna	Abubakar Tafawa Bale...	Bauchi	Education	7390263751	Second Class Upper
471	Albert Martin	30	albertmartin9522@gmail.com	Ekiti	Oyo	Bayero University	Kano	Agriculture	001-839-453-6586x556	Second Class Upper
4788	Alejandro Hill	30	alejandrohill4444@gmail.com	Zamfara	Kaduna	University of Ibadan	Oyo	Education	001-706-424-2930x31099	Pass
4702	Alejandro Jackson	30	alejandrojackson2051@gmail.com	Plateau	Edo	Ahmadu Bello University	Kaduna	Physics	001-027-684-0433x7052	Third Class
9938	Alex Smith	30	alexsmith466@gmail.com	Cross River	Sokoto	University of Nigeria	Enugu	Medicine	388.743.0492x95906	Second Class Upper
8790	Alexander Boyle	30	alexanderboyle6472@gmail.com	Anambra	Osun	Obafemi Awolowo Univ...	Osun	Physics	001-498-673-3249x222	Second Class Lower
3779	Alexander Farrell	30	alexanderfarrell2068@gmail.com	Ekiti	Edo	Abubakar Tafawa Bale...	Bauchi	Mathematics	001-856-392-0413x3794	First Class
5015	Alexander Meyers	30	alexandermeyers2676@gmail.c...	Rivers	Delta	University of Calabar	Cross River	Chemistry	494-111-4553x41175	Third Class
8462	Alexandra Jack...	30	alexandrajackson8384@gmail.c...	Oyo	Sokoto	Abubakar Tafawa Bale...	Bauchi	Economics	208.407.2010	Pass

Answer / interpretation: Youngest age: 20. Oldest age: 30. Returns all tied registrants at either age.

SECTION 5 — Subqueries & Analytical Thinking

Question and answer — use the query in MySQL Workbench

1. Find registrants whose age is above the overall average age.

```
SELECT *
FROM `nysc_registration_data`
WHERE `Age` > (SELECT AVG(`Age`)
               FROM `nysc_registration_data`)
ORDER BY `Age` DESC;
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
8746	Dennis Martin	30	dennismartin1268@gmail.com	Lagos	Sokoto	University of Calabar	Cross River	Mechanical Engineering	(850)760-7414x1316	Third Class
8750	Karen Hart	30	karenhart3704@gmail.com	Sokoto	Edo	Abubakar Tafawa Balewa University	Bauchi	Political Science	(158)743-8918	Second Class Lower
8754	Martin Blake	30	martinblake9873@gmail.com	Plateau	Ekiti	University of Port Harcourt	Rivers	Business Administration	4504711450	First Class
8760	Laura Ferrell	30	lauraferrell1109@gmail.com	Delta	Abuja	University of Calabar	Cross River	Mass Communication	139.865.2696x596	Second Class Lower
8765	Timothy Fowler	30	timothyfowler1080@gmail.com	Enugu	Edo	University of Nigeria	Enugu	Economics	953-672-3951x16259	First Class
8787	Rebecca Smith	30	rebeccasmith5978@gmail.com	Kano	Bauchi	University of Calabar	Cross River	Political Science	182-224-8066	Second Class Upper
8790	Alexander Boyle	30	alexanderboyle6472@gmail.com	Anambra	Osun	Obafemi Awolowo University	Osun	Physics	001-498-673-3249x222	Second Class Lower
8796	Brandy Chen	30	brandychen5395@gmail.com	Osun	Enugu	University of Benin	Edo	Biology	868-212-0007x374	Second Class Upper
8798	Carla Mejia	30	carlamejia572@gmail.com	Abuja	Enugu	Bayero University	Kano	Education	(169)989-4844x5998	Second Class Upper
5839	Nicole Shaw	29	nicoleshaw5433@gmail.com	Lagos	Bauchi	University of Lagos	Lagos	Agriculture	001-755-042-2340x68...	Second Class Lower
5841	Anthony Bean	29	anthonybean1559@gmail.com	Cross River	Plateau	Bayero University	Kano	Biology	001-579-244-4972	Second Class Lower
5859	Derek Wilson	29	derekwilson9836@gmail.com	Zamfara	Rivers	Obafemi Awolowo University	Osun	Mass Communication	(275)499-3918	First Class
5872	Jimmy Bates	29	jimmybates7229@gmail.com	Enugu	Katsina	Abubakar Tafawa Balewa University	Bauchi	Mathematics	733.161.2157x45698	Third Class
5874	Michelle Bailey	29	michellebailey8738@gmail.com	Abuja	Benue	Bayero University	Kano	Agriculture	502.486.0023	Second Class Lower
5885	Lawrence Allen	29	lawrenceallen5307@gmail.com	Plateau	Benue	University of Lagos	Lagos	Electrical Engineering	(596)935-7384x326	Second Class Upper
5906	Frederick Bailey	29	frederickbailey3706@gmail.com	Kano	Abuja	University of Port Harcourt	Rivers	Physics	575.994.9084x6257	Third Class
2203	Randall Bartlett	28	randallbartlett8724@gmail.com	Rivers	Lagos	University of Calabar	Cross River	Mass Communication	9231335199	First Class
2216	Frank Davis	28	frankdavis2323@gmail.com	Katsina	Enugu	University of Ibadan	Oyo	Electrical Engineering	+1-438-581-3090x97439	Pass
2247	Natalie Mack	28	nataliemack4026@gmail.com	Cross River	Kaduna	Obafemi Awolowo University	Osun	Mathematics	+1-695-890-3636	Second Class Upper
2280	Katherine Graham	28	katherinegraham4713@gmail.com	Plateau	Anambra	University of Port Harcourt	Rivers	Education	(636)872-9958	First Class
2288	Mr. Richard Wa...	28	mr.richardwaltersmd3182@gma...	Kaduna	Enugu	Bayero University	Kano	Mathematics	2677002808	Second Class Upper
4950	Laura Mccann	26	lauramccann6462@gmail.com	Kaduna	Oyo	University of Benin	Edo	Biology	9437297709	First Class
4958	Michael Day	26	michaelday7610@gmail.com	Delta	Zamfara	University of Ibadan	Oyo	Medicine	001-840-730-5555x2975	Second Class Upper
4967	Kiara Butler	26	kiarabutler2580@gmail.com	Edo	Kaduna	University of Lagos	Lagos	Medicine	979-296-8607x308	First Class
5014	Luke Schroeder	26	lukeschroeder151@gmail.com	Osun	Ekiti	University of Port Harcourt	Rivers	Agriculture	001-749-794-2105x24...	Third Class
5080	Steven Schmidt	26	stevenschmidt9771@gmail.com	Osun	Plateau	University of Calabar	Cross River	Mechanical Engineering	+1-176-237-3134x0206	Third Class

Answer / interpretation: Returns registrants older than the dataset-wide average (about 24.96 years).

SECTION 5 — Subqueries & Analytical Thinking

Question and answer — use the query in MySQL Workbench

2. List all registrants who attended an institution located in the same state as their origin state.

```
SELECT *
FROM `nysc_registration_data`
WHERE `Origin State` = `Institution State`;
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
5	Eddie Martinez	27	eddiemartinez9172@gmail.com	Rivers	Sokoto	University of Port Harcourt	Rivers	Medicine	839-894-7196x59342	Pass
29	Steven Butler	29	stevenbutler2776@gmail.com	Enugu	Kano	University of Nigeria	Enugu	Chemistry	699.300.2489x4517	Second Class Upper
36	Thomas Rivers	20	thomasrivers9742@gmail.com	Enugu	Benue	University of Nigeria	Enugu	Business Administration	001-415-115-0552x0351	Second Class Upper
38	Daniel Rogers	29	danielrogers6853@gmail.com	Kano	Oyo	Bayero University	Kano	Electrical Engineering	485-291-1894x2622	Second Class Lower
57	Sandra Williams	21	sandrawilliams7850@gmail.com	Rivers	Osun	University of Port Harcourt	Rivers	Computer Science	(373)245-5618	Second Class Lower
77	Kathleen Simon	24	kathleensimon8305@gmail.com	Edo	Anambra	University of Benin	Edo	Electrical Engineering	934.261.9737	Third Class
82	Carla Orozco	26	carlaorozco7519@gmail.com	Kano	Rivers	Bayero University	Kano	Physics	080.862.5911x8243	Second Class Upper
85	Lawrence Hudson	21	lawrencehudson986@gmail.com	Bauchi	Plateau	Abubakar Tafawa Balewa University	Bauchi	Agriculture	685-125-0220x59306	Second Class Lower
94	Jeffrey Ruiz	25	jeffreyruiz4139@gmail.com	Kaduna	Sokoto	Ahmadu Bello University	Kaduna	Electrical Engineering	637.691.4485	Second Class Lower
128	Kevin Washington	30	kevinwashington6474@gmail.com	Lagos	Sokoto	University of Lagos	Lagos	Electrical Engineering	765.601.6922x01630	Pass
149	Caitlin Rodriguez	21	caitlinrodriguez3951@gmail.com	Rivers	Abuja	University of Port Harcourt	Rivers	Biology	869-762-9621x912	Second Class Upper
189	Deborah Powell	29	deborahpowell840@gmail.com	Edo	Abuja	University of Benin	Edo	Medicine	+1-742-035-4106x459	Second Class Lower
214	Caitlin Jackson	21	caitlinjackson9172@gmail.com	Oyo	Anambra	University of Ibadan	Oyo	Physics	(398)308-3787x9955	Second Class Lower
220	Larry Fuller	21	larryfuller7008@gmail.com	Bauchi	Sokoto	Abubakar Tafawa Balewa University	Bauchi	Economics	572.747.5830x5846	Second Class Lower
227	Caitlin Benjamin	21	caitlinbenjamin2022@gmail.com	Lagos	Sokoto	University of Lagos	Lagos	Mechanical Engineering	841-061-6772	Third Class

Answer / interpretation: Returns 484 registrants whose origin state matches their institution state.

SECTION 5 — Subqueries & Analytical Thinking

Question and answer — use the query in MySQL Workbench

3. Identify the most common course of study among registrants.

```
SELECT `Course Study`, COUNT(*) AS `Registrant Count`
FROM `nysc_registration_data`
GROUP BY `Course Study`
ORDER BY `Registrant Count` DESC
LIMIT 1;
```

Course Study	Registrant Count
Biology	711

Answer / interpretation: Result: Biology, 711 registrants.

SECTION 5 — Subqueries & Analytical Thinking

Question and answer — use the query in MySQL Workbench

4. Find the preferred state of the youngest registrant.

```
SELECT `Fullname`, `Age`, `Preferred State`
FROM `nysc_registration_data`
WHERE `Age` = (SELECT MIN(`Age`)
               FROM `nysc_registration_data`)
ORDER BY `Fullname`;
```

Fullname	Age	Preferred State
Amy Collier	20	Anambra
Amy Cruz	20	Anambra
Amy Curtis	20	Delta
Amy Rhodes	20	Enugu
Amy Rodriguez	20	Ekiti
Amy Sanford	20	Delta
Andre Taylor	20	Edo
Andrea Marks	20	Oyo
Andrea Patterson	20	Edo
Andrew Baker MD	20	Kaduna
Andrew Holt	20	Katsina
Andrew Robinson	20	Sokoto
Andrew Rodrig...	20	Borno
Angel Davis	20	Bauchi
Angela Allen	20	Lagos
Angela Fleming	20	Enugu

Answer / interpretation: Youngest age is 20; multiple registrants tie, so the query returns each tied registrant and preferred state.

SECTION 5 — Subqueries & Analytical Thinking

Question and answer — use the query in MySQL Workbench

5. Show the institution(s) that produced the highest number of First Class graduates.

```
WITH first_class_counts AS (
  SELECT `Institution`, COUNT(*) AS `First
Class Count`
  FROM `nysc_registration_data`
  WHERE `Graduating Honors` = 'First Class'
  GROUP BY `Institution`
)
SELECT `Institution`, `First Class Count`
FROM first_class_counts
WHERE `First Class Count` = (
  SELECT MAX(`First Class Count`) FROM
first_class_counts
);
```

Institution	First Class Count
University of Calabar	232

Answer / interpretation: Result: University of Calabar, 232 First Class graduates.

SECTION 5 — Subqueries & Analytical Thinking

Question and answer — use the query in MySQL Workbench

6. List all origin states whose registrants prefer to serve outside their home state.

```sql
SELECT DISTINCT `Origin State`
FROM `nysc_registration_data`
WHERE `Preferred State` != `Origin State`
ORDER BY `Origin State`;
```

Origin State
Abuja
Anambra
Bauchi
Benue
Borno
Cross River
Delta
Edo
Ekiti
Enugu
Kaduna
Kano
Katsina
Kano
Katsina
Lagos
Osun
Oyo
Plateau
Rivers
Sokoto
Zamfara

Answer / interpretation: Returns origin states with at least one registrant whose preferred state differs from their origin state.

SECTION 5 — Subqueries & Analytical Thinking

Question and answer — use the query in MySQL Workbench

7. Determine which institution state contributes the most registrants.

```sql
SELECT `Institution State`, COUNT(*) AS `Registrant Count`
FROM `nysc_registration_data`
GROUP BY `Institution State`
ORDER BY `Registrant Count` DESC
LIMIT 1;
```

Institution State	Registrant Count
Cross River	1049

Answer / interpretation: Result: Cross River, 1,049 registrants.

SECTION 5 — Subqueries & Analytical Thinking

Question and answer — use the query in MySQL Workbench

8. Find all registrants whose course study matches any course studied at University of Lagos (using a subquery).

```
SELECT *
FROM `nysc_registration_data`
WHERE `Course Study` IN (
  SELECT DISTINCT `Course Study`
  FROM `nysc_registration_data`
  WHERE `Institution` = 'University of
Lagos'
);
```

Id	Fullname	Age	Gmail	Origin State	Preferred State	Institution	Institution State	Course Study	Telephone	Graduating Honors
1	Norma Fisher	26	normafisher6891@gmail.com	Kano	Anambra	University of Calabar	Cross River	Economics	876-475-9382x421	Third Class
2	Danielle Browning	24	daniellebrowning7809@gmail.com	Sokoto	Edo	University of Ibadan	Oyo	Political Science	001-924-115-7815x659	Second Class Upper
3	Christine Tran	24	christinetran2290@gmail.com	Rivers	Bauchi	Obafemi Awolowo University	Osun	Biology	840.801.6097x535	Pass
4	Amy Stark	29	amystark2408@gmail.com	Borno	Rivers	Ahmadu Bello University	Kaduna	Biology	328.711.5871	Second Class Lower
5	Eddie Martinez	27	eddiemartinez9172@gmail.com	Rivers	Sokoto	University of Port Harcourt	Rivers	Medicine	839-894-7196x59342	Pass
6	Cody Salazar	30	codysalazar3351@gmail.com	Cross River	Osun	Bayero University	Kano	Chemistry	+1-471-122-0186x84833	Pass
7	Leslie Herring	24	leslieherring1021@gmail.com	Cross River	Lagos	Ahmadu Bello University	Kaduna	Mathematics	001-477-515-9179	Third Class
8	Sarah Sanchez	30	sarahsanchez19@gmail.com	Bauchi	Osun	University of Benin	Edo	Business Administration	001-413-525-6012x3098	Second Class Lower
9	Angela Martinez	21	angelamartinez3131@gmail.com	Edo	Delta	University of Ibadan	Oyo	Physics	991.615.1090	Second Class Upper
10	Wesley Robbins	28	wesleyrobbins7340@gmail.com	Kaduna	Kaduna	University of Benin	Edo	Biology	300.869.1413x1456	Pass
11	Jonathan Williams	27	jonathanwilliams1787@gmail.com	Borno	Cross River	Obafemi Awolowo University	Osun	Mathematics	634-579-2302	First Class
12	Gloria King	28	gloriaking5452@gmail.com	Cross River	Enugu	Abubakar Tafawa Balewa U...	Bauchi	Political Science	+1-720-769-8456x428	Pass
13	Judith Miller	24	judithmiller7291@gmail.com	Kaduna	Bauchi	University of Port Harcourt	Rivers	Medicine	001-084-237-5945x9924	Pass
14	Jason Curtis	23	jasoncurtis4757@gmail.com	Oyo	Enugu	University of Nigeria	Enugu	Computer Science	9352337696	Pass
15	Shane Bradley	30	shanebradley4261@gmail.com	Osun	Kaduna	Ahmadu Bello University	Kaduna	Agriculture	001-027-142-7878x90075	Second Class Upper
16	John Lamb	22	johnlamb634@gmail.com	Kaduna	Cross River	University of Port Harcourt	Rivers	Chemistry	(381)206-6503x00891	Pass

Answer / interpretation: Returns registrants whose course appears among courses studied by registrants at University of Lagos.

BONUS CHALLENGES

Question and answer — use the query in MySQL Workbench

1. Create a query to find the percentage distribution of each graduating honor category.

```
SELECT `Graduating Honors`,
       COUNT(*) AS `Graduate Count`,
       ROUND(100.0 * COUNT(*) / (SELECT COUNT(*)
FROM `nysc_registration_data`), 2) AS `Percentage`
FROM `nysc_registration_data`
GROUP BY `Graduating Honors`
ORDER BY `Percentage` DESC;
```

Graduating Honors	Graduate Count	Percentage
Second Class Upper	2080	20.80
Third Class	2021	20.21
First Class	2020	20.20
Second Class Lower	1973	19.73
Pass	1906	19.06

Answer / interpretation: Percentages: Second Class Upper 20.80%; Third Class 20.21%; First Class 20.20%; Second Class Lower 19.73%; Pass 19.06%.

BONUS CHALLENGES

Question and answer — use the query in MySQL Workbench

2. Find the top 3 most common institutions.

```sql
SELECT `Institution`, COUNT(*) AS `Registrant Count`
FROM `nysc_registration_data`
GROUP BY `Institution`
ORDER BY `Registrant Count` DESC
LIMIT 3;
```

Institution	Registrant Count
University of Calabar	1049
Abubakar Tafawa Balewa University	1038
University of Benin	1029

Answer / interpretation: Top 3: University of Calabar (1,049); Abubakar Tafawa Balewa University (1,038); University of Benin (1,029).

BONUS CHALLENGES

Question and answer — use the query in MySQL Workbench

3. Show for each origin state: total registrants, average age, and most common preferred state.

```
WITH preferred_counts AS (
  SELECT `Origin State`, `Preferred State`, COUNT(*) AS preference_count
  FROM `nysc_registration_data`
  GROUP BY `Origin State`, `Preferred State`
), ranked_preferences AS (
  SELECT `Origin State`, `Preferred State`, preference_count,
         ROW_NUMBER() OVER (
           PARTITION BY `Origin State`
           ORDER BY preference_count DESC, `Preferred State` ASC
         ) AS rn
  FROM preferred_counts
), origin_summary AS (
  SELECT `Origin State`, COUNT(*) AS `Total Registrants`,
         ROUND(AVG(`Age`), 2) AS `Average Age`
  FROM `nysc_registration_data`
  GROUP BY `Origin State`
)
SELECT s.`Origin State`, s.`Total Registrants`, s.`Average Age`,
       r.`Preferred State` AS `Most Common Preferred State`
FROM origin_summary s
LEFT JOIN ranked_preferences r
  ON s.`Origin State` = r.`Origin State` AND r.rn = 1
ORDER BY s.`Origin State`;
```

Origin State	Total Registrants	Average Age	Most Common Preferred State
Abuja	514	24.95	Benue
Anambra	488	25.24	Lagos
Bauchi	517	24.83	Benue
Benue	532	24.86	Kaduna
Borno	541	24.76	Cross River
Cross River	463	24.98	Osun
Delta	455	24.93	Osun
Edo	507	25.08	Cross River
Ekiti	507	25.16	Enugu
Enugu	511	25.00	Oyo
Kaduna	484	25.23	Katsina
Kano	493	25.06	Katsina
Katsina	525	24.91	Anambra
Lagos	495	24.78	Osun
Osun	493	24.86	Enugu
Oyo	486	24.98	Katsina
Plateau	483	25.09	Ekiti
Rivers	491	24.87	Delta
Sokoto	516	24.82	Rivers
Zamfara	499	24.92	Osun

Answer / interpretation: One row per origin state. ROW_NUMBER picks one state in a tie (alphabetically first).

BONUS CHALLENGES

Question and answer — use the query in MySQL Workbench

4. Write a query that finds the difference between the average ages of male and female registrants (if a gender column existed — describe how you'd modify the dataset).

```
    -- The supplied dataset has no Gender column. Add/populate it from a verified source first:
ALTER TABLE `nysc_registration_data` ADD COLUMN `Gender` VARCHAR(20) NULL;

-- After the Gender column is populated accurately:
SELECT
  ROUND(AVG(CASE WHEN `Gender` = 'Male' THEN `Age` END), 2) AS `Average Male Age`,
  ROUND(AVG(CASE WHEN `Gender` = 'Female' THEN `Age` END), 2) AS `Average Female Age`,
  ROUND(AVG(CASE WHEN `Gender` = 'Male' THEN `Age` END)
      - AVG(CASE WHEN `Gender` = 'Female' THEN `Age` END), 2) AS `Male Minus Female Average Age`
FROM `nysc_registration_data`;
```

Answer / interpretation: No gender result can be calculated from the supplied data. Add the column and populate it from a reliable source; do not guess values.

BONUS CHALLENGES

Question and answer — use the query in MySQL Workbench

5. Using window functions (if supported), rank registrants by age within each institution.

```
SELECT `Institution`, `Fullname`, `Age`,
       RANK() OVER (
         PARTITION BY `Institution`
         ORDER BY `Age` DESC
       ) AS `Age Rank Within Institution`
FROM `nysc_registration_data`
ORDER BY `Institution`, `Age Rank Within Institution`,
`Fullname`;
```

Institution	Fullname	Age	Age Rank Within Institution
Abubakar Tafawa Balewa University	Alan Hunter	30	1
Abubakar Tafawa Balewa University	Alexander Farrell	30	1
Abubakar Tafawa Balewa University	Alexandra Jackson	30	1
Abubakar Tafawa Balewa University	Amanda Griffin	30	1
Abubakar Tafawa Balewa University	Alexander Jones	29	82
Abubakar Tafawa Balewa University	Ana Smith	29	82
Abubakar Tafawa Balewa University	Andrea Parker	29	82
Abubakar Tafawa Balewa University	Andrea Smith	29	82
Abubakar Tafawa Balewa University	Andrew Ross	29	82
Abubakar Tafawa Balewa University	Andrew Yoder	29	82
Bayero University	Regina Crawford	29	85
Bayero University	Rita Vargas	29	85
Bayero University	Robert Allen	29	85
Bayero University	Robert Brown	29	85
Bayero University	Christopher Salas	27	274
Bayero University	Christopher Sutton	27	274
Bayero University	Curtis Ross	27	274
Bayero University	Cynthia Guerra	27	274

Answer / interpretation: Ranks registrants oldest-to-youngest within each institution; ties receive the same rank.

Project wrap-up

Validate each result by executing the corresponding SQL query.

- Dataset size: 10,000 records and 11 columns.
- Approximate Average age: 25years; youngest age: 20; oldest age: 30.
- Most common course: Biology (711 registrants).
- Most represented institution: University of Calabar (1,049 registrants).
- Most represented institution state: Cross River (1,049 registrants).
- Rivers-origin registrants preferring Lagos: 24.
- Important limitation: the supplied dataset has no Gender column, so the male/female average-age comparison requires verified gender data before it can be calculated.
- SQL compatibility: the bonus queries using CTEs/window functions require MySQL 8.0 or later.

Thank You

NYSC Registration Data Analysis with MySQL

Questions & Discussion

Prepared using the supplied NYSC registration dataset and SQL practice questions